

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 2

Mark scheme

AQA spec 3.1.2.3.4; The living environment; textbook pp. 50-71

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them (1) • Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists (1) • Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) (1) • Endemics have small ranges and populations so are vulnerable to extinction (1)	4
01.2	<i>1 mark for each point.</i> • The birds evolved without mammal predators so nest on the ground / in burrows with no defence (1) • Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) (1) • After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor (1)	3
01.3	<i>1 mark for each point.</i> • Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria (1) • Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range (1)	2
01.4	<i>1 mark.</i> • Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
02.1	<i>1 mark for each point.</i> - Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) - Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) - Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3
02.2	<i>1 mark for each point.</i> - Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) - Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) - Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) - Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
02.3	<i>1 mark for each point.</i> - Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) - Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) - Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	
03.1	<i>1 mark for each point.</i> - Upwelling currents bring nutrients from the seabed to surface waters (1) 24-hour daylight in summer allows a huge phytoplankton bloom (1) - Algae also grow under sea ice where krill shelter and feed; krill support most of the food web (1)	3
03.2	<i>1 mark for each point.</i> - Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web (1) - Toothfish overfishing removes the food of sperm whales and elephant seals (1) - Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch) (1)	3
03.3	<i>1 mark for each point.</i> - Ice on land stores 70% of the Earth's fresh water, keeping sea levels low (1) - High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) (1) - Carbon sequestration: carbon in algae eaten by krill sinks to the seabed in krill faeces (1)	3

Question	Marking guidance	Marks
03.4	<i>1 mark.</i> Stronger winds spreading ice; more snowfall from warmer, moister air freezing onto slush; fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	
04.1	<i>1 mark for each point.</i> - Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected); sea ice loss removes krill habitat and lowers albedo (1) - Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) (1) - Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution; visitor numbers rising rapidly at a few landing sites (1) - Reserves of coal, oil, gold, uranium, platinum etc; pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult (1)	4
04.2	<i>1 mark for each point.</i> - Treaty signed by 52 countries including all active there; no military activity (except supporting research), no nuclear explosions or radioactive waste dumping (1) - No commercial mineral exploitation (1) - Fisheries regulated by CCAMLR: krill and some fish exploited, fisheries closed if over-exploited; no commercial exploitation of mammals (1) - Tourism control: no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions (1) - Waste management: all wastes removed (fuel containers, batteries, plastics, oil); sewage only discharged where sea currents disperse it; old waste removed unless removal causes more damage (1) - Only long-term residents are research scientists and support staff; strict regulation of research impacts (1)	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
05.1	<i>Correct answer only.</i> B (1)	1
05.2	<i>1 mark for each point.</i> - Rainforest: high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification) (1) - Antarctica: very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life (1) - Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom; rainforest supports over half of terrestrial species on 10% of land (1) - Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes; Antarctic species adapted to extreme fluctuation (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
06.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Zooxanthellae are photosynthetic algae living inside the polyp cells (1) • The algae absorb CO₂ and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) (1) • They supply the polyp with oxygen and glucose - over 95% of its food energy (1) • In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water (1) • Polyps also feed on plankton captured by cilia and nematocysts (1)	4
06.2	<i>1 mark for each point.</i> • Corals deposit a skeleton of calcium carbonate (calcite) (1) • Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink (1)	2
06.3	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income • Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs • Justified conclusion	9
	Model answer - Level 3 (9 marks)	

Question	Marking guidance	Marks
	Tropical coral reefs cover about one per cent of the oceans yet support over a quarter of marine species, a diversity built on the symbiosis between polyps and zooxanthellae that recycles nutrients in nutrient-poor water. They matter to people for fisheries of fish, lobsters and conch; for medicines such as AZT, Ara-C, dolastatin, secosteroids, bryostatin and trabectedin discovered in reef organisms; as a carbon sink, since every carbonate unit in coral skeleton absorbs a molecule of carbon dioxide; for dissipating wave energy that would erode low-lying islands; and for tourism income.	
	They are also unusually vulnerable because they need narrow abiotic conditions. Anchors, divers, litter and coral extraction kill polyps; ornaments are collected; sediment from rivers and coastal development smothers them; oil, sewage and nutrients cause algal shading; ocean acidification hinders skeleton formation; warming causes bleaching; overfishing of giant tritons lets crown-of-thorns starfish destroy reefs; introduced lionfish eat native fish; and loss of mangroves and seagrass removes the nurseries and sediment traps reefs depend on.	
	Marine protected areas such as the Great Barrier Reef, Belize and Chagos combine mooring buoys, diver education, no-take zones that raise catches outside them, sustainable tourism and protection of linked habitats, while fishing controls set closed seasons, minimum and maximum sizes and ban dynamite, trawling and small mesh. These work well against local threats where they are enforced, and no-take zones have measurable benefits, but they cannot touch warming and acidification, which need international action, enforcement in remote areas is weak, and restrictions conflict with local livelihoods. Local conservation buys time; only global climate action can secure the reefs.	
	Weaker answer - Level 1 (3 marks)	
	Coral reefs are important because lots of fish live there and people catch them for food. Tourists visit reefs to dive and this brings money. Reefs also protect the coast from waves. Reefs are threatened by climate change which causes bleaching, and by pollution and damage from boats. They are conserved by making marine protected areas where fishing is banned. This helps but climate change is still a problem.	
	What makes the difference	
	• Level 1 has the right headings but Level 3 explains each (why reefs are diverse, why they are a carbon sink, how each threat acts).	
	• Level 3 uses named reserves and specific management measures; Level 1 gives one generic measure.	
	• Level 3's evaluation separates local from global threats and reaches a nuanced judgement; Level 1 states that climate change is 'still a problem'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	